**MSc Taxonomy, Biodiversity and Evolution - Project Proposal Academic Year**

| | |
|---|---|
| **Student's Name:** Indi Rick (CID 06068014) | **Project handing in: End-August** |
| **Supervisor's Name (+email):** Max Barclay (m.barclay@nhm.ac.uk) | **Location of Work:** N/A |

**Project 2nd Marker/Examiner Name (+email):** Dmitry Telnov (d.telnov@nhm.ac.uk)

**Supervisor: Agreement to complete all marking associated with the project *within given set timeframe***

**Supervisor: details of any planned time spent away & name of covering supervisor**

**Project Title:** Predictive analysis of the invasion of *Chrysolina americana* in the UK.

**Outline of Work:**

Research question & hypotheses:

-**Question**: How are the invasion dynamics of *C. americana* mediated by climate warming trends and urban density across the UK?

-**H1**: *C. americana*'s spread is shaped by thermal constraints, on two fronts: both global climate warming and the urban heat island effect accelerate their proliferation around the metropolitan hubs.

-**H2**: Higher human density causes earlier colonisation of areas: jump dispersal is enabled by the inter-city plant trade and the presence of gardens, allowing the beetle to bypass the limitations of diffusion dispersal.

Datasets used:

1-Royal Horticultural Society's Rosemary beetle (*Chrysolina americana*) monitoring. (link)
2-Met Office's Annual Average Temperature Change - Projections (12km). (link)
3-UKCEH's UK gridded population at 1km resolution for 2021 based on Census 2021/2022. (link)
4-Black-headed Cardinal Beetle (*Pyrochroa coccinea*) occurrences in the UK, on GBIF. (link)

Workflow:

-**Data preparation**: climate forecast being the data with the least resolution (12km grid), datapoints within the boundaries of a grid cell will be integrated to it, so that each cell contains values for: 1. the number of beetles, 2. the total human population and 3. the temperature increase change for different warming scenarios (+1.5, +2.0, +2.5, +3.0 and +4.0°C). Spatial data wrangling will be performed in ArcGIS Pro.

-**Current and predictive distribution analyses**: the rate of colonisation of *C. americana* (km/year) will be calculated using a Haversine formula. Using a negative binomial generalised linear model (NBGLM) in R, the occurrence data will be combined with the predictive variables (thermal limit and urban density) to forecast *C. americana's* range across the UK, according to the different climate projections.

-**Comparison with other organisms, variables, and models**: we will compare the findings with a native species, *P. coccinea,* to calibrate the expansion due to temperature without the urban density factor. Precipitation will also be tested as a secondary variable, to determine its impact. Finally, we will compare the results with other species distribution models, like Maximum Entropy (machine learning method).

-**Data visualisation**: in ArcGIS Storymaps, cartographic storytelling of historical & predicted distributions.

**Special Requirements (eg. equipment, chemicals). *Note to supervisor: expenses ~£500 (up to £1K) can be given as reimbursement for molecular projects:*** N/A

| | | | |
|---|---|---|---|
| **Collections to be Used** | N/A | **Approved by** | N/A |
| **Signature of Student** | Indi Rick | **Date** | 08/05/2026 |
| **Signature of Supervisor** | M.V.L. BARCLAY | **Date** | 8/5/2026 |
| **Approved by** | | **Date** | |